

Quaid-e-Azam Solar Power (Pvt) Ltd

Project for IT Officer Position

Web Development + AI Interview Assignment

AI-Powered Tax Software Product

1. Assignment Overview

You are required to build a web application for a fictional tax software company.

The application should help users choose the right tax software product based on their tax situation.

This task is inspired by modern tax software websites. You are not expected to know tax law. All required product rules and recommendation rules are provided in this document.

The purpose of this task is to evaluate your:

- frontend development skills
- backend/API understanding
- UI/UX thinking
- ability to build reusable components
- ability to implement business logic
- ability to work with structured product data
- AI integration or AI simulation skills
- code organization and documentation

Automated testing is not mandatory. You only need to document manual verification in the README. Automated tests will be treated as a bonus.

2. Goal

Build an AI-assisted product recommendation website where users can:

1. View available tax software products
2. Compare product features
3. Answer a recommendation questionnaire
4. Get a recommended product
5. Ask an AI assistant product-selection questions
6. View product rules/configuration from an admin page

3. Recommended Tech Stack

You may use your preferred stack, but the recommended stack is:

Frontend

- React or Next.js

- TypeScript
- Tailwind CSS, CSS Modules, ShadCN, Material UI, or similar

Backend

You may use:

- Next.js API routes
- Express.js
- NestJS
- FastAPI
- or any similar backend framework

AI

You may use:

- OpenAI API
- Gemini API
- Anthropic API
- or a simulated/mock AI assistant

If you do not have an AI API key, you may build a simulated AI assistant. The simulation should still behave like a real product recommendation assistant.

4. Required Pages

You must build the following pages:

1. Landing page
2. Products page
3. Product comparison page
4. Recommendation wizard
5. AI assistant page or widget
6. Admin/config page

5. Products

Use the following fictional products.

Product 1: Free

Price

CAD \$0

Best for

Users with a simple personal tax situation.

Supports

- salary income
- student income

- basic personal return
- simple tax slips

Does not support

- medical expenses
- donations
- investment income
- rental income
- self-employment income
- business income
- expert help

Product 2: Deluxe

Price

CAD \$30

Best for

Users with common deductions and expenses.

Supports

- everything in Free
- medical expenses
- donations
- employment expenses
- family-related deductions

Does not support

- investment income
- rental income
- self-employment income
- corporate tax filing

Product 3: Premier

Price

CAD \$50

Best for

Users with investments, rental income, capital gains, or foreign income.

Supports

- everything in Deluxe
- investment income
- capital gains
- foreign income
- rental income

Does not support

- self-employment income
- corporate tax filing

Product 4: Self-Employed

Price

CAD \$70

Best for

Freelancers, contractors, gig workers, and sole proprietors.

Supports

- salary income
- student income
- medical expenses
- donations
- investment income
- rental income
- freelance income
- gig-work income
- business expenses
- home-office expenses
- vehicle expenses

Does not support

- incorporated business filing

Product 5: Expert Assist

Price

CAD \$120

Best for

Users who want to file themselves but need help from a tax expert.

Supports

- all personal tax situations
- expert chat
- expert video call
- expert review before filing

Does not support

- full handoff filing
- incorporated business filing

Product 6: Expert Full Service

Price

CAD \$250

Best for

Users who want an expert to prepare and file their return.

Supports

- all personal tax situations
- document upload
- expert prepares return
- expert files return
- progress tracking

Does not support

- incorporated business filing

Product 7: Business Corporate

Price

CAD \$400

Best for

Incorporated companies.

Supports

- corporate tax return
- business revenue
- business expenses
- corporate filing review

Does not support

- personal tax return

Product 8: Nil Corporate Return

Price

CAD \$175

Best for

Incorporated companies with no revenue.

Supports

- incorporated company filing
- no-revenue company
- nil return

Does not support

- personal tax return
- companies with revenue

6. Product Data Requirement

Products must be stored in a structured data file or database.

Do not hardcode all product information directly inside UI components.

Example:

```
export type Product = {  
  id: string;  
  name: string;  
  price: number;  
  currency: "CAD";  
  description: string;  
  bestFor: string[];  
  supports: {  
    salaryIncome: boolean;  
    studentIncome: boolean;  
    medicalExpenses: boolean;  
    donations: boolean;  
    employmentExpenses: boolean;  
    investmentIncome: boolean;  
    capitalGains: boolean;  
    foreignIncome: boolean;  
    rentalIncome: boolean;  
    freelanceIncome: boolean;  
    gigWorkIncome: boolean;  
    businessExpenses: boolean;  
    homeOfficeExpenses: boolean;  
    vehicleExpenses: boolean;  
    expertHelp: boolean;  
    fullService: boolean;  
    corporateFiling: boolean;  
    nilCorporateReturn: boolean;  
  };  
};
```

You may modify or improve this structure if needed.

Important: recommendation logic should be separate from UI components.

7. User Questionnaire

Build a multi-step questionnaire.

The wizard should include:

- progress indicator
- next/back buttons
- validation
- result screen
- restart option

Step 1: Filing Type

Question:

What are you filing for?

Options:

1. Personal return
2. Freelancer / self-employed
3. Incorporated company

Step 2: Income Sources

Question:

Which income sources apply to you?

Options:

- Salary income
- Student income
- Investment income
- Capital gains
- Rental income
- Freelance income
- Gig-work income
- Business revenue
- Foreign income

User may select multiple options.

Step 3: Deductions / Expenses

Question:

Which deductions or expenses apply to you?

Options:

- Medical expenses
- Donations
- Employment expenses
- Home-office expenses
- Vehicle expenses
- Business expenses
- No special deductions

User may select multiple options.

Step 4: Help Preference

Question:

How much help do you want?

Options:

1. I want to file myself
2. I want expert help while filing
3. I want an expert to file for me

Step 5: Company Revenue

Show this step only if the user selected Incorporated company.

Question:

Did the company have revenue?

Options:

1. Yes, company had revenue
2. No, company had no revenue

Step 6: Result

Show:

- recommended product
- price
- reason for recommendation
- matched user answers
- optional upgrade if applicable
- warnings if applicable
- disclaimer
- restart button

Example result:

Recommended product: Self-Employed

Reason: You selected freelance income and home-office expenses. The Self-Employed product supports freelance income and business-related expenses.

Disclaimer: This recommendation is for product guidance only and is not tax, legal, or financial advice.

8. Recommendation Rules

Use this priority order:

1. Incorporated company
2. Expert Full Service
3. Expert Assist
4. Self-Employed
5. Premier
6. Deluxe
7. Free

Higher-priority rules override lower-priority rules.

Rule 1: Incorporated Company

If the user selects Incorporated company:

- Recommend Business Corporate

If the incorporated company had no revenue:

- Recommend Nil Corporate Return

This rule overrides all personal tax products.

Rule 2: Expert Full Service

If the user selects:

“I want an expert to file for me”

Recommend:

- Expert Full Service

Exception:

- If the user selected Incorporated company, use Business Corporate or Nil Corporate Return.

Rule 3: Expert Assist

If the user selects:

“I want expert help while filing”

Recommend:

- Expert Assist

Exception:

- If the user selected Incorporated company, use Business Corporate or Nil Corporate Return.

Rule 4: Self-Employed

Recommend Self-Employed if the user selected any of these:

- Freelancer / self-employed filing type
- freelance income
- gig-work income
- business revenue as an individual
- business expenses
- home-office expenses
- vehicle expenses

Rule 5: Premier

Recommend Premier if the user selected any of these:

- investment income
- capital gains
- rental income

- foreign income

Rule 6: Deluxe

Recommend Deluxe if the user selected any of these:

- medical expenses
- donations
- employment expenses

Rule 7: Free

Recommend Free if the user has a simple situation:

- personal return
- salary income and/or student income
- no special deductions
- no investments
- no rental income
- no freelance income
- no expert help requested

9. Recommendation Output

The recommendation engine should return a structured result.

Example:

```
export type RecommendationResult = {
  recommendedProductId: string;
  recommendedProductName: string;
  price: number;
  confidence: "low" | "medium" | "high";
  reasons: string[];
  matchedInputs: string[];
  optionalUpgrade?: {
    productId: string;
    productName: string;
    reason: string;
  };
  warnings?: string[];
  disclaimer: string;
};
```

Example:

```
{
  "recommendedProductId": "self-employed",
  "recommendedProductName": "Self-Employed",
  "price": 70,
  "confidence": "high",
  "reasons": [
    "You selected freelance income.",
    "You selected home-office expenses.",
    "Self-Employed supports freelance income and business-related expenses."
  ],
  "matchedInputs": ["freelanceIncome", "homeOfficeExpenses"],
```

```
"disclaimer": "This recommendation provides general product guidance only and is not tax, legal, or financial advice."
}
```

10. Page Details

10.1 Landing Page

Route example:

/

Include:

- header
- hero section
- CTA: “Find my product”
- CTA: “Compare products”
- product preview cards
- how it works section
- FAQ section
- footer

The design should be clean, modern, and responsive.

10.2 Products Page

Route example:

/products

Show all products.

Each product card should include:

- product name
- price
- short description
- best for
- key supported features
- CTA button

Bonus:

- filter by product type
- sort by price
- search by feature

10.3 Product Comparison Page

Route example:

/compare

Create a comparison table.

Compare products by:

- price
- salary income
- donations
- medical expenses
- investment income
- rental income
- freelance income
- business expenses
- expert help
- full service
- corporate filing
- nil return

The table should be readable on mobile.

10.4 Recommendation Wizard

Route example:

/recommend

Build the questionnaire described above.

Required:

- progress indicator
- step navigation
- validation
- recommendation result

10.5 AI Assistant

Route example:

/assistant

Build an AI assistant that answers product-selection questions.

Example questions:

1. "I have salary income and donations. Which product should I use?"
2. "I am a freelancer with home-office expenses. Can I use Free?"
3. "I own an incorporated company with no revenue. What should I choose?"
4. "I have investment income and rental income. Which product fits me?"
5. "What is the difference between Premier and Self-Employed?"
6. "I want someone else to file for me. What should I select?"

The assistant must answer based on product data and rules.

10.6 Admin / Config Page

Route example:

/admin/products

Build a simple admin/config page.

Minimum requirement:

Show product configuration in a readable admin view.

Display:

- product ID
- product name
- price
- best for
- supported features
- unsupported features

The admin page can be read-only.

Bonus:

- edit product data locally
- validate product schema
- export product config as JSON
- show missing/invalid fields

This page is included to check whether the candidate can structure data cleanly and display configuration-driven content.

11. AI Assistant Requirements

The AI assistant can be real or simulated.

Option A: Real AI

Use an AI API.

The AI should receive:

- user question
- product data
- recommendation rules
- safety instructions

The AI should return a structured response.

Example:

```
{
  "answer": "Based on the provided product rules, Premier appears suitable because it supports investment income and rental income.",
  "recommendedProduct": "Premier",
  "confidence": "high",
  "reasons": [
    "User mentioned investment income.",
    "User mentioned rental income.",
    "Premier supports both."
  ],
}
```

```
"disclaimer": "This is general product guidance only and is not tax, legal, or financial advice."  
}
```

Option B: Simulated AI

If you do not have an API key, build a simulated assistant.

The simulated assistant should:

- accept natural language input
- detect keywords
- map user question to product rules
- return a helpful answer
- include disclaimer
- avoid unsupported claims

12. AI Safety Rules

The assistant must not provide real tax, legal, financial, or accounting advice.

The assistant must not say:

- “You are guaranteed a refund.”
- “You definitely qualify for this deduction.”
- “The tax authority will accept your return.”
- “This is legal advice.”
- “This is professional tax advice.”
- “You must use this product.”

The assistant should say:

- “Based on the provided product rules...”
- “This product appears suitable...”
- “Please consult a qualified tax professional for tax advice.”
- “This is general product guidance only.”

Example unsafe user prompt:

“Can you guarantee I will get a refund?”

Expected response:

“I cannot guarantee refunds or tax outcomes. I can only provide general product guidance based on the product rules.”

13. Suggested API Routes

You may create these API routes or equivalent server-side functions.

GET /api/products

Returns all products.

POST /api/recommend

Accepts questionnaire answers and returns a recommendation.

POST /api/assistant

Accepts a user question and returns an AI or simulated assistant response.

14. Validation Requirements

Minimum validation:

- filing type is required
- at least one income source is required
- help preference is required
- company revenue answer is required if filing type is incorporated company

Handle contradictions where possible.

Examples:

- “No special deductions” selected with “Medical expenses”
- Incorporated company selected but personal-only product recommended
- User asks AI to invent a feature

15. Testing Requirements

Automated testing is not mandatory.

However, your README must include a short Manual Verification section explaining how you checked the main flows.

At minimum, manually verify these scenarios:

Scenario	Expected Result
Salary only	Free
Salary + donations	Deluxe
Investment income	Premier
Rental income	Premier
Freelance income	Self-Employed
Home-office expenses	Self-Employed
Wants expert help	Expert Assist
Wants expert to file	Expert Full Service
Incorporated company with revenue	Business Corporate
Incorporated company with no revenue	Nil Corporate Return
AI asked about refund guarantee	Safe disclaimer response

Automated tests are optional and will be treated as a bonus.

16. Optional Bonus Features

These are not required.

- automated tests for recommendation engine
- editable admin product config
- real AI API integration
- localStorage wizard persistence
- product filters
- dark mode
- deployment
- PDF export of recommendation
- multilingual support
- better accessibility
- CI workflow

17. README Requirements

Your README must include:

1. Project name
2. Short description
3. Tech stack
4. Setup instructions
5. How to run the app
6. Environment variable instructions, if AI API is used
7. Routes/pages implemented
8. Product data structure explanation
9. Recommendation engine explanation
10. AI assistant explanation
11. Admin/config page explanation
12. Manual verification section
13. Known limitations
14. Future improvements
15. AI usage disclosure

AI Usage Disclosure

Include a section called:

Use of AI During Development

Mention:

- which AI tools you used
- what you used them for
- what code or design was AI-assisted
- how you reviewed and verified the output
- what you wrote manually

18. Deliverables

Submit:

1. GitHub repository link
2. Live deployed link, if available
3. README
4. Screenshots or short demo video, if possible
5. Any assumptions you made

Deployment is optional but preferred.

Suggested deployment platforms:

- Vercel
- Netlify
- Render
- Railway
- Fly.io

19. Time Expectation

Recommended time:

- 1 week for full assignment

If you cannot complete everything, prioritize:

1. product data model
2. recommendation engine
3. wizard
4. result page
5. tests
6. AI assistant

A smaller but clean and well-tested submission is better than a large incomplete submission.

20. Evaluation Rubric

Total: 100 points

Area	Points
Requirement coverage	15
UI/UX and responsiveness	15
Products and comparison pages	10
Recommendation wizard	15
Recommendation engine correctness	20
AI assistant and safety behavior	15
Admin/config page	5
Code quality and architecture	5

Area	Points
Optional automated tests / bonus quality	Bonus

21. What We Will Evaluate

Requirement Coverage

We will check:

- required pages exist
- products are displayed correctly
- wizard works
- result page works
- AI assistant works or is simulated
- admin/config page exists
- README is complete

UI/UX

We will check:

- clean layout
- mobile responsiveness
- readable content
- clear buttons
- usable forms
- clear recommendation result

Recommendation Engine

We will check:

- rule priority is correct
- logic is separated from UI
- results include reasons
- edge cases are handled
- corporate rules override personal rules
- expert help rules override lower plans

AI Assistant

We will check:

- AI is grounded in product data
- assistant does not invent unsupported features
- assistant includes disclaimer
- assistant avoids tax/legal/financial advice
- unsafe questions are handled correctly

Admin / Config Page

We will check:

- product data is displayed clearly
- supported and unsupported features are visible
- data-driven configuration is easy to understand
- page demonstrates clean product-data modeling

Code Quality

We will check:

- readable structure
- reusable components
- good TypeScript usage
- clean naming
- minimal duplication
- maintainable business logic

22. Disallowed Shortcuts

Avoid these:

1. Do not build only static pages with no working logic.
2. Do not hardcode all recommendation logic inside JSX.
3. Do not ignore the priority rules.
4. Do not create a generic chatbot disconnected from product data.
5. Do not expose API keys in frontend code.
6. Do not skip basic validation.
7. Do not submit code that cannot run.
8. Do not copy a real website design exactly.
9. Do not provide real tax/legal/financial advice through the assistant.
10. Do not skip the admin/config page.

23. Final Submission Checklist

Before submitting, confirm:

- ☐ App runs locally
- ☐ README is complete
- ☐ Products are stored in structured data
- ☐ Recommendation logic is separated from UI
- ☐ Landing page works
- ☐ Products page works
- ☐ Comparison page works
- ☐ Wizard works
- ☐ Result page shows correct recommendation
- ☐ AI assistant works or is simulated
- ☐ AI assistant includes disclaimer
- ☐ Admin/config page exists
- ☐ Manual verification is documented
- ☐ GitHub repository is accessible

24. Final Note

We value clean, working software more than unnecessary complexity.

A smaller but well-structured and functional implementation is better than a large incomplete project.

Focus on:

- clean UI
- correct logic
- readable code
- maintainable structure
- useful AI behavior
- admin/config visibility
- clear documentation

Good luck.